

Model Context Protocol (MCP) Cheat Sheet

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Baseline: MCP spec 2026-07-28; Python mcp 2.2.0 (3.12) and fastmcp 4.0.10; Spring Boot 4.1.1 + Spring AI 2.0.1 (Java 21; MCP Java SDK 2.0 speaks only 2025-11-25); Node.js 24 LTS with @modelcontextprotocol/server 2.2.0; MCP Inspector 2.8.0. Verified 2026-09-29. Quick reference only — verify against the official docs before production use.

2026-07-28 AT A GLANCE

- **Stateless core:** no initialize, no `Mcp-Session-Id`; each request carries version, client info and capabilities in `_meta`; optional `server/discover`
- **MRTTR:** `resultType complete | input_required` replaces `server-to-client requests` (elicitation)
- **Headers:** `Mcp-Method` + `Mcp-Name` required on Streamable HTTP; `x-mcp-header` custom headers
- **Caching:** `ttlMs`, `cacheScope` on `list/read` results; one `subscriptions/listen stream`
- **Extensions:** `Tasks` (`io.modelcontextprotocol/tasks`), MCP Apps, Skills over MCP
- **Auth:** RFC 9207 `iss` check; CIMD replaces DCR (deprecated)
- **Deprecated:** `roots`, `sampling`, `logging`, `HTTP+SSE` (12-month minimum window); removed: `ping`, `logging/setLevel`

PRIMITIVES

Primitive	Controlled by	Methods	docs-assistant
Tool	model	<code>tools/list</code> , <code>tools/call</code>	<code>search_docs</code>
Resource	application	<code>resources/list</code> , <code>resources/read</code> , <code>templates</code>	<code>docs://{component}/{page}</code>
Prompt	user	<code>prompts/list</code> , <code>prompts/get</code>	<code>answer_with_citations</code>
Elicitation	server asks user	<code>input_required</code> (MRTTR)	which component?

Tool annotations (`readOnlyHint`, `destructiveHint`, `idempotentHint`, `openWorldHint`) are untrusted hints. Schemas: JSON Schema 2020-12.

TRANSPORTS

- **stdio:** local child process; one JSON-RPC message per line on `stdin/stdout`; log to `stderr` only
- **Streamable HTTP:** one endpoint (e.g. `/mcp`); POST per request; reply JSON or SSE stream; Accept both types
- **Headers:** `MCP-Protocol-Version`, `Mcp-Method`, `Mcp-Name`; must match body (`HeaderMismatch`)
- **Validate** `Origin/Host`; bind `127.0.0.1` locally
- Any instance can serve any request: plain round-robin, no sticky sessions; multi-instance `requestState` needs a shared key
- **HTTP+SSE:** deprecated — use Streamable HTTP
- Java (MCP Java SDK 2.0 / Spring AI 2.0.1) speaks 2025-11-25 only: no `server/discover`, no MRTTR; STATELESS mode needs no session affinity
- **Deploy:** container per stack, TLS, rate limit on `Mcp-Method/Mcp-Name`, MCP Registry (preview) `server.json`

SERVER SKELETONS (DOCS-ASSISTANT)

Python — mcp 2.2 MCPServer

```
from mcp.server import MCPServer
mcp = MCPServer("docs-assistant")
@mcp.tool()
def search_docs(query: str, top_k: int = 3) -> SearchResult: ...
mcp.run("streamable-http") # or "stdio"
```

Java — Spring AI 2.0 (spring-ai-starter-mcp-server-webmvc)

```
spring.ai.mcp.server.protocol=STREAMABLE|STATELESS|SSE
@mcpTool(name="search_docs", generateOutputSchema=true)
public SearchResult searchDocs(@McpToolParam(description="q") String query) { }
```

TypeScript — @modelcontextprotocol/server 2.2

```
const s = new McpServer({name:"docs-assistant",version:"1.0.0"});
s.registerTool("search_docs", {inputSchema: z.object({query: z.string()}),
  outputSchema}, async ({query}) => ({content:[...], structuredContent}));
serveStdio(() => s) // HTTP: createMcpHandler + Express
```

TOOL DESIGN RULES

- Few, task-shaped tools; not one per endpoint
- **Description:** when to use, args, what next
- Concise default, detail on request, cursors
- Readable IDs; actionable `isError` text
- Curated beats OpenAPI-generated
- Tool tokens $T = \sum_i (d_i + s_i)$ (description + schema): measure

RAG OVER MCP

- **Tools:** `search_docs` (`outputSchema`) + `read_passages` (`result_set`, `ranks`); `resource docs://{...}`; `prompt answer_with_citations`
- Host LLM writes the answer — no sampling, server makes no model call
- Handles (random, short TTL, keyed by verified user) replace sessions
- **ACL** filter from the token's identity, pre-filtered in pgvector / Spring AI VectorStore, never a tool argument

MESSAGE SHAPES (JSON-RPC 2.0)

Every request: version, client, capabilities in `_meta`

```
POST /mcp  Mcp-Method: tools/call  Mcp-Name: search_docs
MCP-Protocol-Version: 2026-07-28
Accept: application/json, text/event-stream
{"jsonrpc":"2.0","id":3,"method":"tools/call",
 "params":{"name":"search_docs",
  "arguments":{"query":"matrix rank","component":"algebra"},
  "meta":{"io.modelcontextprotocol/protocolVersion":"2026-07-28",
    "io.modelcontextprotocol/clientInfo":{"name":"c","version":"1"},
    "io.modelcontextprotocol/clientCapabilities":{}}}}
```

Result: structuredContent validated by outputSchema

```
{"result":{"resultType":"complete","isError":false,
 "content":[{"type":"text","text":"..."}],
 "structuredContent":{"hits":[...]}}
```

Elicitation round trip (MRTTR)

```
- {"resultType":"input_required",
  "inputRequests":{"key":{"method":"elicitation/create",...},
  "requestState":"<opaque>"}
-> retry same call + {"inputResponses":{"key":{...},
  "requestState":"<echoed>"}
```

- `x-mcp-header` on a tool parameter mirrors it as `Mcp-Param-<name>` (string, integer, boolean only)
- `server/discover` → `supportedVersions`, `capabilities`, `serverInfo` in `result_meta`
- **Tool failure:** `isError: true` in the result; protocol error: JSON-RPC error (-32602 bad params/unknown resource; -32020 HeaderMismatch, -32021 and -32022 (defined in the reserved range))

AUTH CHECKLIST

- OAuth 2.1, PKCE S256; server = resource server
- 401 + WWW-Authenticate resource_metadata; PRM (RFC 9728) at `/.well-known/oauth-protected-resource`
- AS metadata RFC 8414; resource param (RFC 8707)
- CIMD (not DCR); validate `iss` (RFC 9207)
- Validate signature, issuer, expiry, audience, scope on every request
- Never pass the client token downstream

INSPECTOR CLI 2.8

```
mcp-inspector --cli URL
--transport http
--protocol-era modern|legacy
--method tools/call
--tool-name search_docs
--tool-args-json '{"...}'
--format json

Web: mcp-inspector; TUI: --tui; dev
loop uv run mcp dev server.py. Exit:
0 ok, 3 auth, 4 unreachable, 5 tool error.
Use --protocol-era legacy for Java.
Elicitation: test with curl.
```

HOST CONFIG

Claude Code

```
claude mcp add --transport http --scope project docs URL
claude mcp add --transport stdio docs -- python server.py
# --scope project writes .mcp.json; /mcp = OAuth login
```

VS Code / Copilot — .vscode/mcp.json

```
{"servers":{"docs":{
  "type":"http","url":"http://127.0.0.1:8000/mcp"}}}
```

Copilot cloud agent (tools only, no OAuth remotes)

```
{"mcpServers":{"docs":{"type":"local","command":"python","args":
["server.py"],
  "env":{"K":"$COPILOT_MCP_K"},"tools":["search_docs"]}}}
```

SECURITY CHECKLIST

- **Tokens:** validate audience; no passthrough; exchange or use own credential downstream
- **Proxies:** per-client consent; exact `redirect_uri`; single-use state
- **SSRF:** HTTPS only, no auto-redirects, egress proxy, size/time limits
- **Local servers:** sandbox; show exact command; validate `Origin/Host`
- **Model-facing text:** pin tool descriptions; results are data; no private data + untrusted input + outbound channel together
- **Destructive tools:** own scope, approval gate, reversible, audited; least privilege